

A429X Technical Status Report

Oct 8, 2026 · @Shubham

A429X specification rev 1.1 implements the v1.0 feature set except 12 deferred requirements, and 104 of its 127 requirements are fully verified. The core runs on an AMD ZCU106 board at 100 and 250 MHz. What remains is testing on real ARINC 429 wiring, support for other FPGA vendors and certification evidence.

At a glance

Every simulation, gate-level and board test of rev 1.1 passes, and the core meets its 250 MHz target both out of context and on the board.

Item	Status (2026-10-08)
Specification	A429X-PS-001 rev 1.1
Requirements (RTM draft 2.0)	104 of 127 fully verified; 5 partial; 12 not implemented (deferred); 6 process; 0 implemented without a test
RTL evidence baseline	git cec1f8d (spec rev 1.1)
Simulation regression	14 benches + 3 clock-variant runs: 17 of 17 pass, 0 assertion messages (1,370 s wall)
Gate-level simulation	24 of 24 checks pass on the rev 1.1 netlist
Code coverage, 14 benches	Statement 95.35 %, branch 80.78 %, condition 76.77 %, toggle 31.4 %
Timing, product core	250 MHz met after place and route out of context on UltraScale+ -2: WNS +0.048 ns
Timing, ZCU106 builds	100 MHz: WNS +1.037 ns. 250 MHz: WNS +0.014 ns
Area, eval build (4 Tx + 4 Rx, all options, line emulator)	53,844 LUT, 44,559 FF, 28 RAMB36 + 10 RAMB18, 55 DSP
Area, product build (no line emulator)	52,005 LUT, 43,614 FF, 35 DSP

Item	Status (2026-10-08)
Board, rev 1.1	Self-test H0-H14: 47 of 47. Rate test: 36 of 36 at 100 and at 250 MHz. Acceptance tests: 50 of 50 in two runs, 49 of 50 in one (a statistical miss, see Hardware validation)
Repository	Branch dev/lhm-ram-sva-hcmp, 35 commits ahead of main, not pushed; latest commit ba2a0e4

Product scope

A429X is an ARINC 429 Part 1 controller with two options built on top: the Sealed Label Protocol (SLP) for authentication, and the Line Health Monitor (LHM) with a physical fingerprint intrusion detector.

Feature group	Implemented in rev 1.1
Baseline core	1-16 Tx and 1-16 Rx channels at 100 kbps, 12-14.5 kbps or a custom rate; fractional NCO bit-rate generator; odd, even and transparent parity; Tx FIFO, 64- or 256-entry periodic scheduler and timed transmit; 1024-entry label/SDI filter; Rx FIFO with 64-bit timestamps; per-label mailboxes and refresh monitor; parity, length, gap, rate, glitch and line-state errors; automatic rate detection; bus load; 7 error-injection codes (test mode only); AXI4-Lite control
Sealed Label Protocol	Shared AES-128-CMAC engine; up to 4 seal groups per Tx channel; 21-, 42- or 63-bit tags; 48-bit freshness value sent as 13 bits; MONITOR, EARLY and ENFORCE delivery; sync seals with optional time; fail-secure sealing; 64-entry security event log
Key management	16 write-only 128-bit key slots; SP 800-108 key derivation; zeroize by register or pin; power-up known-answer test with lockout; key access only on the secure bus

Feature group	Implemented in rev 1.1
Line Health Monitor	FE0 and FE1 front ends; per-word metrics with min, max, mean and variance; limits, WARN/ALARM health score, RATE_PPM; one engine shared by all Rx channels
Fingerprint IDS	Learns a mean (4 fractional bits) and MAD per feature; per-word flag on the largest feature term (D_MAX, default) or the sum (D_SUM); window alarm; drift adaptation
Tx readback	Wrap-around compare, lost-word and foreign-transmitter detection
Safety	SECDED state codes in all 35 FSMs, encoder watchdog, receiver activity timeout, off-bus internal loopback, crypto self-test
Lab add-on	Virtual Line Emulator: 4 analog line models with slew, noise, ringing, glitches and dropouts, plus contention, gap-injection, MITM and replay attack modes

Licence tiers and area. Out-of-context synthesis on Vivado 2025.2, XCZU7EV-2, measured on the 2026-10-06 RTL. All tiers meet 100 MHz. Later revisions changed the full build by under 2 %.

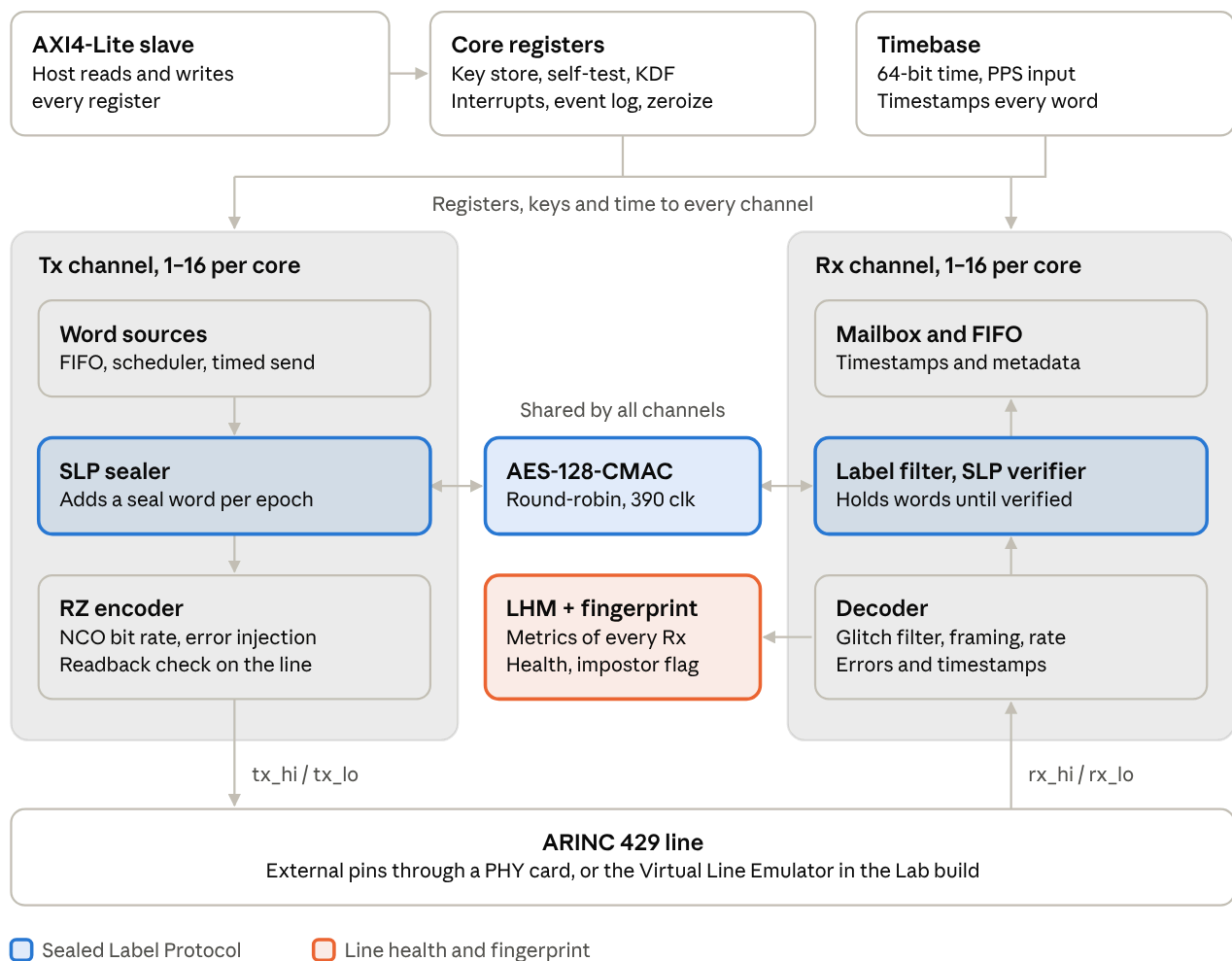
Tier	Contents	LUT
Core, 1 Tx + 1 Rx	Baseline ARINC 429 core	3,977
Core, 4 + 4	Baseline ARINC 429 core	18,080
Sense, 4 + 4	Core + LHM + fingerprint + readback	28,593
Secure, 4 + 4	Core + SLP + key store	40,793
Complete, 4 + 4	All features	51,129
Lab add-on	Virtual Line Emulator, 4 lines	+1,839 (20 DSP)

Main build parameters. `G_CLK_HZ` (20–250 MHz), `G_NUM_TX` and `G_NUM_RX` (1–16), FIFO depths, `G_SCHED_ENTRIES` (0, 64, 256), `G_MAILBOX` (none, 256, 1024), `G_AUTH_EN`, `G_SECURE_BUS`, `G_LHM_TIER` (none, FE0, FE1), `G_FP_EN`, `G_RB_EN` and `G_VLE_EN`. The specification also lists `G_AXIS_EN`, `G_ECC_EN` and `G_CFG_PROT`, which belong to deferred features.

Architecture

The core is one AXI4-Lite slave with up to 16 Tx and 16 Rx channels, which share a single AES-128-CMAC engine and a single LHM and fingerprint engine. Sharing the LHM engine alone saves about 3.4k LUTs per Rx channel.

Every channel shares one crypto engine and one line-health engine



A429X core · Tx and Rx data paths, shared engines, line

Transmit words flow down from the host to the line, and received words flow up from the line to the host. The sealer and the verifier queue their CMAC jobs on the shared engine, and every decoder feeds its line metrics to the shared LHM engine. Every FSM state register in these blocks is a SECDDED code word, so a single-bit upset is corrected on the next clock.

Development history

The specification went from a first draft to rev 1.1 in four days, 2026-10-05 to 2026-10-08. Each revision after 0.4 was driven by a board or verification finding, and each was re-verified in simulation and on the ZCU106.

Rev	Date	Main change	Commit
1.1	2026-10-08	Tx readback limits registered: 250 MHz margin up from +0.008 to +0.048 ns. ZT-04 scores false positives over both halves of a case. Nine specification corrections from the customer-document review. Yosys maps the core to Lattice ECP5	cec1f8d
1.0	2026-10-08	Fingerprint mean keeps 4 fractional bits, which ends the run-to-run spread of detection under 50 mV of noise. <code>FP_INVSIG</code> clamp fixed. 250 MHz ZCU106 build. Bit rate (TX-004) verified on silicon at 100 and 250 MHz	27349f3
0.9	2026-10-07	Bare-metal driver implements the whole §14 API, with the interrupt through the GIC. Reference design built from the IP-XACT package. PS DDR set for the lab board's x16 SODIMM	99073e5, e01b88e, bc740f9
0.8	2026-10-07	Product core meets 250 MHz after place and route (about 170 MHz before): LHM statistics pipeline, registered fingerprint operands, mailbox scanner, rate classifier and FV base	55b2a5f

Rev	Date	Main change	Commit
0.7	2026-10-07	D_MAX becomes the default fingerprint distance (FP_CTRL[7] is now D_SUM)	3595414
0.6	2026-10-06	SECDED state codes in every FSM with defined recovery, encoder watchdog and receiver activity timeout (SAF-001, SAF-003)	ee1a962
0.5	2026-10-06	Fixes for the two board findings: timing errors delivered as flags (R12) and the D_MAX distance (R11). Fingerprint term pipelined. Python RZ coder and OpenSSL cross-check	f71fe3d
0.4	2026-10-05	Verification closure: seven new benches found about 20 RTL defects, all fixed. Constant multiplies as shift-add (–20 DSP). First board self-test on this RTL (2026-10-06): 32 of 32	401d187, b24962e
0.3	2026-10-05	Security hardening: zeroization scope, fail-secure sealing, secure-bus default, seal check order	2d4e4f9
0.1–0.2	2026-10-05	First full draft, then aligned with the v1.0 RTL	1f52e44 (main), b968724

Verification

The rev 1.1 RTL passes all 17 regression runs with 0 assertion messages (2026-10-08, 11:35–11:58), with the same check counts as rev 1.0.

Environment

- **Simulator:** Vivado xsim 2025.2. Embedded A429X_ASSERT checks run on every clock in every bench; the same properties exist as concurrent assertions for formal tools.
- **Independent reference model:** model/pya429x.py is a bit-exact encoder, decoder, sealer and verifier built on the Python cryptography package and cross-checked against OpenSSL. model/lhm_model.py models the LHM and fingerprint arithmetic.

- **Driver in the loop:** `tb_a429x_drv` compiles the C driver into the simulation through DPI-C and runs every §14 call against the RTL.
- **Scripts:** `sim/regress.ps1` runs everything in parallel; `sim/coverage.ps1` merges coverage; `sim/lint_verilator.ps1` runs the Verilator lint.

Regression results (rev 1.1)

Bench	What it checks	Checks
<code>tb_a429x_soak</code>	4 Tx + 4 Rx under random traffic and error injection, 2 seal groups; an upset storm of 1,590 single-bit upsets into all 35 FSMs, all corrected	2,392
<code>tb_a429x_slp_tx</code>	Sealer bit-exact against the Python model: tag widths, epoch sizes, sync types, FV wrap and exhaustion, fail-secure blocking, legacy-receiver transparency	388
<code>tb_a429x_slp_rx</code>	Verifier: sync and time checks, quarantine per mode, alarm and fingerprint policy, replay, zeroize, event log	301
<code>tb_a429x_lhm</code>	Every LHM metric against a pin-level reference; fingerprint learn, restore, detect, adapt and alarm against the model	230
<code>tb_a429x</code>	Full eval build over AXI4-Lite: timing, integrity, attacks, fingerprint, degradation, zeroize, key derivation, FSM upset recovery (T0-T19)	212
<code>tb_a429x_tx</code>	Scheduler, gap, FIFO, timed release, parity, AXI channel ordering, every single-bit upset of the Tx FSMs, encoder watchdog	212
<code>tb_a429x_rx</code>	Latency, timestamps, framing, glitches, mailbox, auto-rate, PPS, loss-of-bus timeout	190
<code>tb_a429x_equiv</code>	Full and core builds side by side: SLP off leaves ARINC 429 behaviour unchanged	159
<code>tb_a429x_sec</code>	Secure-bus build: KAT lockout, malformed seals, FV window edges, zeroize by register and pin	104

Bench	What it checks	Checks
tb_a429x_drv	The C driver against the RTL, including the interrupt handler	68
tb_a429x_phy	Injection codes, rate skew, glitch widths, auto-rate switching, refresh monitor	46
tb_a429x_min	Core tier, 1 Tx + 1 Rx	20
tb_a429x_lhm_t1	FEO-only LHM tier	16
tb_cmac	FIPS-197 AES and RFC 4493 CMAC vectors, two requesters, zeroize mid-job	All pass
Soak at 20 MHz	Soak traffic at the 20 MHz clock floor	2,394
Rx at 20 MHz	Clock-sensitive Rx subset at 20 MHz	54
Tx clock sweep	Bit rate and edge placement at 20, 33.3, 100, 150 and 250 MHz; worst rate error 0.61 ppm	5 of 5
tb_gls	Gate-level netlist of the rev 1.1 RTL: KAT, loopback, SLP golden	24

Static checks. Verilator 5.050 -Wall passes with reviewed waivers. Vivado lint gives 56 messages and 0 errors, each rule justified in `verif/lint_cdc_waivers.md`. The routed ZCU106 design gives 8 CDC-3 information entries and 0 unsafe crossings. The MISRA C:2012 check (Cppcheck addon) finds 0 Mandatory or Required violations.

Code coverage (rev 1.1)

Metric	14-bench merge	LHM bench alone (6-receiver variant)
Statement	95.35 %	97.0 %
Branch	80.78 %	89.2 %
Condition	76.77 %	90.1 %
Toggle	31.4 %	12.8 %

Toggle coverage is low because directed tests never fully toggle the wide data paths: 48-bit FVs, 64-bit timestamps, 128-bit keys and RAM data. The specification's goal of 100 % with justified exclusions is still open.

Defects found. The benches found and fixed about 30 RTL defects. Examples: a crypto RAM read one cycle early (visible only against the independent model), zeroize missing the CMAC subkeys and tag, a protected word sent unsealed, stale data after an AXI upset, and the integer fingerprint mean behind the noisy-line detection spread.

Requirements traceability

Of 127 requirements, 104 are fully verified and no implemented requirement lacks a test. The matrix (`verif/A429X_RTM.md` , draft 2.0) is generated from `verif/rtm_mapping.json` against the rev 1.1 evidence baseline cec1f8d.

Area	Simulation	Assertion	Analysis	Partial	Not implemented	Process	Total
GEN (general)	3			1		1	5
TX (transmitter)	14	2					16
RX (receiver)	20						20
BT (bulk transfer)					3		3
SLP (sealing)	23		1		1	3	28
KEY (keys)	6		1				7
LHM (line health)	9				3		12
FP (fingerprint)	4			1			5
RB (readback)	3				1		4
TIME (timebase)	4						4
IF (interface)	1		1				2
IRQ (interrupts)	3						3
CLK (clock, reset)	2		1		1		4
SAF (safety)	4				1	2	7
SW (software)			1	2	2		5
ZCU (reference design)			1	1			2
All	96	2	6	5	12	6	127

103 requirements have at least one executed simulation or assertion check. Eight are covered only by analysis, inspection or software tests: GEN-005, SLP-003, KEY-002, IF-001, CLK-003, SW-003, SW-004 and ZCU-001. No requirement is waiting on a board run.

The five partial requirements

ID	Priority	Gap	What closes it
FP-001	P1	Targets met on the board on a clean line and at 23 mV of emulated noise. At 50 mV, eight rev 1.0/1.1 runs flag 87–94 % (FE0) and 80–98 % (FE1) of attacker words with 0 false positives: steady, but below the 99 % target. No test with real oscillators, cabling and measured noise	Lab ZT-04 with real hardware (Phase C)
GEN-005	P3	Synthesis done with Vivado, Efinity (map only) and Yosys for UltraScale+ and Lattice ECP5 (no place and route). Microchip and Intel never tried	Libero and Quartus runs; place and route on a second vendor
SW-001	P3	MISRA checked with Cppcheck's addon, which is not a qualified tool	A qualified MISRA C:2012 checker
ZCU-002	P3	Each channel selects only the line emulator or the external pins; no PMOD-loopback / PHY-card selection	Add the mux to the board top, or document both as external pins
SW-003	P3	<code>pya429x</code> is not yet a cocotb scoreboard: Icarus cannot build the IP and Windows cocotb has no Verilator support	cocotb with Verilator on a Linux host

Timing and resources

The product core has met its 250 MHz target since rev 0.8, and rev 1.1 widened the margin to +0.048 ns. All figures are Vivado 2025.2 on XCZU7EV-2.

Product core at 250 MHz (out of context, place and route, no line emulator)

Rev	Result at 4 ns	Limiting path
1.1	Met, WNS +0.048 ns, WHS +0.014 ns, 0 of 165,178 endpoints failing	Timebase to Rx word timestamp (88 % route)
1.0	Met, WNS +0.008 ns	Tx readback lost-word limit (a product of two configuration values in one clock)
0.8	Met, WNS +0.017 ns	Quarantine RAM read enable, decoder timestamp
0.5–0.7	About 170 MHz (synthesis), 656–660 endpoints failing	LHM statistics variance update
v1.0	About 158 MHz (synthesis)	Fingerprint distance accumulator

ZCU106 builds (rev 1.1, full Zynq UltraScale+ block design)

Build	WNS / WHS	LUT	FF	BRAM	DSP	Worst path
100 MHz, line emulator on	+1.037 / +0.010 ns	53,811	45,482	33	55	Inside the line emulator
250 MHz, line emulator off	+0.014 / +0.006 ns	52,635	44,682	33	35	Core reset synchronizer to event-log register (97 % route)

The 250 MHz build needs the `Performance_ExplorePostRoutePhys0pt` strategy. With the default strategy a rev 0.9 build failed 175 endpoints by up to 0.153 ns.

Resources by block (rev 1.1, out-of-context synthesis, each instance)

Block	LUT	FF	BRAM36	DSP
Rx channel (2 seal groups, 1024 mailbox, FIFO 256)	5,966	5,157	7	4
Shared LHM and fingerprint engine (all 4 Rx)	6,320	5,094	0	3
Tx channel (64-entry scheduler, sealer, readback)	4,185	3,109	0	4
Shared AES-128-CMAC engine	1,808	994	10 × RAMB18	0
Core registers, key store, KAT, KDF, event log	2,365	4,046	0	0

Block	LUT	FF	BRAM36	DSP
Timebase, AXI4-Lite slave, top glue	901	359	0	0
Virtual Line Emulator, 4 lines (Lab only)	1,839	945	0	20
Eval build, 4 Tx + 4 Rx, all options	53,844	44,559	28 + 10 RAMB18	55
Product build, same without the line emulator	52,005	43,614	28 + 10 RAMB18	35

The eval build uses 23 % of the device's LUTs and 9 % of its flip-flops. The first build was 81.7k LUTs. Moving LHM storage to LUT-RAM, a single key-store write port and one shared LHM engine cut it to 49.7k. Security and verification fixes then added about 4k, and shift-add constant multiplies saved 20 DSPs. The v1.1 area target for a 4 + 4 Secure + Sense build is 35k LUTs or less.

Performance figures

- **CMAC:** 390 clocks for the longest MAC (budget 600).
- **Worst crypto queue:** 16 Tx + 16 Rx channels requesting at once finish within 12,480 clocks (125 µs at 100 MHz), less than one high-speed word time (360 µs).
- **ENFORCE release after the last tag word:** 0.85 µs for a 4-word epoch, 4.25 µs for a 63-word epoch, at 100 MHz.
- **Measured bit-rate error on silicon:** 27 ppb (HS) and 162 ppb (LS) at 100 MHz; 44 ppb and 631 ppb at 250 MHz. The limit is 0.01 % (100,000 ppb).

Hardware validation on the ZCU106

Every revision since 0.4 has run on an AMD ZCU106 board, and rev 1.1 passes its self-test, rate test and acceptance tests at 100 and 250 MHz. All of this is Phase A: the built-in line emulator models the analog wire, so no external ARINC 429 hardware has been used yet.

Set-up. The ZCU106 (XCZU7EV) runs standalone A53 programs loaded over JTAG by `sw/zcu106_selftest/run_on_board.tcl`, with results on the USB-UART console. The lab board's SODIMM uses x16 devices (Micron MTA4ATF51264HZ-2G6E1, read from its SPD), so builds use `build.tcl -tclargs sodimm_x16`; with the kit's default x8 setting, DDR addresses alias. The external channel pins go to PMOD J55 and J87, but no loopback or PHY card is fitted.

Board programs

Program	Checks	What it covers
<code>sw/zcu106_selftest</code> (H0–H14)	47	Identity and crypto self-test; loopback; the emulated line path; seals bit-exact against the Python model; injection, replay and stripping attacks; a +200 ppm fingerprint attacker; slew and amplitude degradation; zeroize; key derivation and fail-secure blocking; key invalidation; safety monitors (H13); the driver API with the interrupt taken through the GIC (H14)
<code>sw/zcu106_zt</code> (ZT-04, ZT-05, ZT-01)	50	ZT-04: fingerprint statistics with a ± 300 ppm attacker injecting into gaps, on a clean line and under 23 and 50 mV RMS of noise, FE0 and FE1. ZT-05: slew, amplitude and combined degradation sweeps. ZT-01 (scaled): 10^6 words at HS on four channels, 10^4 at LS and 12,000 in auto-rate
<code>sw/zcu106_rate</code>	36	Bit rate and edge placement (TX-004) from receiver timestamps and LHM edge metrics
<code>sw/zcu106_ddrtest</code> , <code>sw/zcu106_spd</code>	—	DDR walking-address and dense-data test over 4 GB; SODIMM identification

Results by revision

Rev	Date	Self-test	Acceptance tests	Other
1.1	2026-10-08	47 of 47	50 of 50 in two runs, 49 of 50 in one	Rate test 36 of 36 at 100 and 250 MHz
1.0	2026-10-08	47 of 47	50 of 50 in five runs	Rate test 36 of 36 at 100 MHz and, for the first time, at 250 MHz

Rev	Date	Self-test	Acceptance tests	Other
0.9 (driver, rev 0.8 RTL)	2026-10-07	47 of 47: polled, through the GIC, from the IP-XACT build, from a clean clone and linked into DDR	50 of 50, also from the IP-XACT build and from DDR	DDR test passes over 4 GB
0.8	2026-10-07	36 of 36, two runs	50 of 50, two runs	
0.7	2026-10-07	36 of 36, two runs	50 of 50, two runs	
0.6	2026-10-06	36 of 36	51 of 52, two runs	The miss: a statistical false positive at <code>D_WORD_TH 5</code>
0.5	2026-10-06	32 of 32	52 of 52	Confirms the R11 and R12 fixes
0.4	2026-10-06	32 of 32	29 of 29	First acceptance tests; found R11 and R12

The rev 1.1 miss was statistical. One legitimate word reached $D = 7$ in the half of a 23 mV case scored at `D_WORD_TH 6`, so that half could not prove a rate below 10^{-4} (it needs 0 in 29,957 words). ZT-04 now scores both halves together: about 60,000 words, where one false positive still proves the rate at 95 % confidence. The next run passed 50 of 50.

Fingerprint results (ZT-04, `D_WORD_TH 6`)

Condition	Attacker words flagged	False positives
Clean line, both front ends	613 of 613	0 in 30,037 genuine words
23 mV RMS noise, <code>D_MAX</code>	99.0–100 %	2 in 480,592 words on rev 1.0–1.1 (about $4 \cdot 10^{-6}$)
50 mV, FE0, rev 0.5–0.9 (10 runs)	42–98 %	0–2 per case

Condition	Attacker words flagged	False positives
50 mV, FE1, rev 0.5–0.9 (10 runs)	25–98 %	0–2 per case
50 mV, FE0, rev 1.0–1.1 (8 runs)	87–94 %	0
50 mV, FE1, rev 1.0–1.1 (8 runs)	80–98 %	0

The 50 mV spread had one cause: the RTL used only the integer part of the learned mean. The genuine span mean sits at 31000 – 0.0015 clock, so it truncated to 30999 or 31000 at random, and a 30999 baseline inflated the spread estimate and cut detection to 45–67 %. Rev 1.0 keeps 4 fractional bits and the spread is gone. Detection under 50 mV is still below the 99 % target at threshold 6; threshold 5 gives 97–99 % with up to 7 false positives per 30,037 words on FE1.

Line health (ZT-05). The LHM warning came before the first damaged word in all three sweeps:

Sweep	WARN	First RATE_ERR	First lost word
Slew	2.5 μ s	5.0 μ s	8.0 μ s
Amplitude	0.90	0.85	0.50
Both	2.0 μ s / 0.95	3.0 μ s / 0.90	8.0 μ s / 0.60

Board findings that changed the design

- **R11:** fingerprint false positives under line noise. Fixed by D_MAX (rev 0.5, default since 0.7) and the fractional mean (rev 1.0).
- **R12:** the rate check discarded valid words on a degraded line. Timing errors are now delivered as flags unless TIMING_STRICT is set (rev 0.5).
- **Driver:** a full Tx FIFO now returns an error, and a429x_tx_send_wait() was added.
- **DDR:** the x16 SODIMM configuration (rev 0.9).
- **Test sequencing:** a receiver enabled before its loopback listened to floating PMOD pins.

Software, models and tools

The bare-metal C driver implements the whole §14 API and runs both in simulation (through DPI-C) and on the board. A Linux kernel driver is still deferred.

Bare-metal driver `sw/a429x`

Group	Calls
Initialization and configuration	<code>a429x_init</code> , <code>a429x_tx_cfg</code> , <code>a429x_rx_cfg</code> , <code>a429x_label_cfg</code>
Transmit	<code>a429x_tx_send</code> , <code>a429x_tx_send_at</code> , <code>a429x_sched_set</code> , <code>a429x_tx_send_wait</code> , <code>a429x_tx_wait_idle</code>
Receive	<code>a429x_rx_read</code> , <code>a429x_mbx_read</code> , <code>a429x_rx_flush</code>
Security	<code>a429x_key_load</code> , <code>a429x_key_derive</code> , <code>a429x_key_invalidate</code> , <code>a429x_zeroize</code> , <code>a429x_slp_tx_cfg</code> , <code>a429x_slp_rx_cfg</code> , <code>a429x_fv_get</code> , <code>a429x_tx_protect</code>
Line health and fingerprint	<code>a429x_lhm_get</code> , <code>a429x_lhm_mean</code> , <code>a429x_fp_baseline_save</code> , <code>a429x_fp_baseline_load</code> , <code>a429x_fp_start</code>
Interrupts and time	<code>a429x_isr</code> , <code>a429x_time_get</code>

The driver uses no dynamic memory. Every call except `a429x_init` and `a429x_isr` takes a `const` device pointer, and configuration calls range-check every field and return `-1` without writing anything. The MISRA C:2012 check (Cppcheck addon, `sw/a429x/misra_check.ps1`) finds 0 Mandatory or Required violations; Advisory rules 2.5, 8.7 and 11.4 are deviated with reasons in `sw/a429x/MISRA.md`.

Other software

- **Linux (`sw/linux`)**: a UIO user-space library, a command-line tool and a device-tree fragment. The generated node is `a429x_0: a429x@a0000000`, compatible `xlnx,a429x-1.0`, interrupt SPI 89 (GIC ID 121).

- **Python reference model (`model/pya429x.py`)**: bit-exact RZ line coder, encoder, decoder, sealer and verifier, plus golden-vector generation (Appendix A format). Companions: `lhm_model.py` (LHM and fingerprint arithmetic), `fp_noise_study.py` (noise and threshold study) and `slp_xcheck_openssl.py` (OpenSSL cross-check run before every regression).
- **Profile tool (`tools/a429x_profile.py`)**: reads an SLP profile, checks seal-label conflicts, computes bus load, worst-case latency and tag strength, and writes C init tables. It has unit tests and was reviewed against the specification in rev 1.0.
- **IP package (`fpga/ip`)**: IP-XACT package for Vivado IP Integrator, XDC and SDC constraint templates, and `check_constraints.tcl` , which passes on the routed ZCU106 design.

Toolchain

Tool	Version	Use
AMD Vivado, xsim, xcrg	2025.2	Synthesis, implementation, simulation, coverage, lint, CDC
AMD Vitis (standalone BSP)	2025.2	Board platform and A53 programs
Verilator (MSYS2)	5.050	<code>-Wall</code> lint of the full RTL
Cppcheck with MISRA addon	2.21	MISRA C:2012 check of the driver
Yosys with sv2v	0.66 / 0.0.13	Open-source synthesis for UltraScale+ and Lattice ECP5
Efinix Efinity	2026.1	Titanium Ti375 synthesis (map stage)
Icarus Verilog, cocotb	13 / 2.1	Installed; Icarus cannot build the IP
Python with <code>cryptography</code> , <code>PyYAML</code>	3.x	Reference model, profile tool, scripts

All tools are installed per user, because the development PC has no administrator rights.

Deviations, limitations and deferred features

Twelve requirements are deferred to the roadmap, a few register features are reserved or partial, and the remaining limitations are documented in the specification (§19, §20.2).

Deferred requirements (not implemented)

ID	Feature	Note
BT-001, BT-002	Bulk Transfer Assist (block transmit, Rx capture window)	Needs the AXI4-Stream ports, also deferred
BT-003, SW-005	ARINC 429 Part 3 (Williamsburg) protocol library	Depends on the Bulk Transfer Assist
SLP-004	Ascon authentication option	Roadmap v1.1
LHM-002	FE2 ADC front end	Roadmap v1.1
CLK-001	ADC clock-crossing FIFO and ADC_RATE	Deferred with FE2; the core stays single-clock
LHM-022	LHM trend ring buffer	Its address region reads 0
LHM-032	Default LHM limit presets	All limits reset to off
RB-004	LHM metrics on the readback path	The readback region has counters only
SAF-002	ECC and TMR options	<code>G_ECC_EN</code> and <code>G_CFG_PROT</code> do not exist yet
SW-002	Linux kernel driver	A UIO user-space library ships instead

Other features not in the v1.0 RTL

- AES key wrap: the UNWRAP command returns `CMD_ERR`.
- The `K_RATE` sliding-window alarm.
- `TIME_CTRL.PPS_EDGE : pps_in` is taken on its rising edge only.
- `TX_IRQ_STATUS` bits 10-12 (`LHM_WARN`, `LHM_ALARM`, `BULK_DONE`) are never set.
- `GCTRL.EN` and `FREEZE_ALL` are reserved.

Partial behaviour

- `GCTRL.LOCK` locks `GCTRL` only; the full configuration lock is planned for v1.1.
- Only the core register block generates `SLVERR`.
- The optional memory BIST is not implemented; internal loopback and the crypto self-test are.

- The mailbox refresh scanner needs a clock of at least 41 MHz to cover 1024 entries in 100 μ s.

Known limitations

- **250 MHz margin is small:** +0.048 ns out of context and +0.014 ns in the ZCU106 block design. A design that places the core at 250 MHz next to other logic may need floorplanning.
- **Fingerprint under heavy noise:** at 50 mV RMS, detection at the recommended threshold is 80–98 %, below the 99 % target. Real line noise has not been measured.
- **Baseline compatibility:** since rev 1.0, `FP_MU` holds $16 \times$ the mean. A fingerprint baseline saved from an earlier revision must be multiplied by 16 before it is restored.
- **Tool workarounds:** xsim mis-evaluates concurrent assertions in later generate instances, so simulation uses immediate assertions. `xelab -debug off` miscomputes AES, so every build uses `-debug typical`. xcrg often stops while writing the HTML coverage report, so scores are read from its dashboard.

Residual risks (specification §19)

- **R1:** prior art and freedom to operate need a professional patent search.
- **R4:** legacy receivers may behave unexpectedly with the seal label; only interoperability tests on real equipment (ZT-30) can show this.
- **R5:** system key management is outside the IP.
- **R6:** replay after a receiver reboot is a documented residual risk; the time-checked sync option mitigates it.
- **R9:** no side-channel (DPA) protection of the keys.

Open work and next steps

The next steps that need only the ZCU106 are the rest of Phase A, a demo and an evaluation package. Real-wire testing and certification need a partner, a lab or outside reviewers.

With the ZCU106 only

Item	Purpose	Status
ZT-02: 24-hour sealing soak in ENFORCE	Phase A: 0 false seal failures over 24 h	Not started

Item	Purpose	Status
ZT-03: the remaining "modify" attack through the line emulator	Phase A: completes the attack suite (inject, replay and strip already pass)	Not started
ZT-01 at the full 10^8 words	Phase A: long loopback, a few hours of board time	10^6 done
ZT-06: zeroize pin on a board push-button	Phase A: tests the real pin path (tied off today)	Not started
Evaluation package	Time-limit logic, per-customer build ID, IEEE 1735 encrypted RTL and simulation model, encrypted ZCU106 bitstream	Not started
Demo recording	Spoofed-word rejection and the line-health warning, for partners and funding applications	Not started
One-page brief	Updated to the rev 1.1 figures (104 of 127, 17 of 17 runs, 250 MHz on silicon, 50 mV detection)	Done (2026-10-08)

Verification closure in the repository

Item	Closes	Needs
Coverage to 100 % with justified exclusions	§15.2 goal	Directed tests and exclusions, mainly toggle
Microchip and Intel synthesis; place and route outside AMD	GEN-005	Libero, Quartus or nextpnr (a download, to be approved)
Qualified MISRA checker	SW-001	A licensed tool
cocotb scoreboard with Verilator	SW-003	A Linux host
Pin mux or documentation	ZCU-002	A decision: build the mux, or document both sources as external pins

Item	Closes	Needs
Deferred features	12 not-implemented requirements	A decision per feature: build for v1.0 or move to v1.1

Needs outside help

Item	Closes	Needs
Phase B: external PMOD loopback (ZT-10)	Real I/O, synchronizers, constraints	8 jumper wires
Phase C: PHY card and cable fixture (ZT-20 to ZT-22, lab ZT-04)	FP-001; electrical compliance	A PHY card (to design) or a partner's ARINC 429 bench
Phase D: interoperability (ZT-30 to ZT-32)	Backward compatibility of sealing (R4)	A commercial ARINC 429 analyzer or a legacy box, via a partner or a rented DRDO facility
DO-254 certification kit	Certification-ready deliverable	PHAC, verification plan and results, independent verification
HDL language decision	Specification open question 1	Ask the first customers (VHDL is common in Indian defence labs)

Business prerequisites (owner: project lead)

- Confirm who owns the IP (the developer or the employer), and set up the company that will sell it.
- File a provisional patent application before any public demonstration.
- Prepare an NDA and an evaluation licence agreement before handing out any evaluation material.
- Obtain an export classification for the Secure tier before any sale abroad.

Evidence and references

Every figure in this report comes from the repository at commit ba2a0e4 (branch dev/lhm-ram-sva-hcmp); the RTL evidence baseline is cec1f8d.

What	Where
Product specification, rev 1.1	A429X_Product_Specification.md (results in §20.3, area in §13.2, risks in §19)
Requirements traceability matrix, draft 2.0	verif/A429X_RTM.md , generated from verif/rtm_mapping.json by verif/rtm_extract.py
Lint, CDC and Verilator waivers	verif/lint_cdc_waivers.md , verif/verilator_lint.vlt
RTL (16 source files)	rtl/
Test benches and golden vectors	tb/
Regression and coverage	sim/regress.ps1 , sim/coverage.ps1 , sim/coverage/modules_summary.md , sim/coverage/modules_summary_lhm.md
Out-of-context synthesis and 250 MHz probe	fpga/ooc_synth.tcl , fpga/ooc_fmax.tcl
ZCU106 reference design	fpga/zcu106/build.tcl (options sodimm_x16 , clk250), fpga/zcu106/a429x_zcu106_top.v
IP-XACT package and constraints	fpga/ip/
Board programs and logs	sw/zcu106_selftest , sw/zcu106_zt , sw/zcu106_rate and their results/ folders (rev 1.1 logs are named *_rev11_*)
Driver and MISRA record	sw/a429x/ , sw/a429x/MISRA.md
Linux user-space library	sw/linux/
Python models	model/
Profile tool	tools/a429x_profile.py , tools/test_a429x_profile.py

What	Where
Configuration management plan	plans/A429X_Hardware_Configuration_Management_Plan.md
Customer documents (issue 3)	docs/A429X_Technical_Reference.pdf , docs/A429X_Brochure.pdf , docs/A429X_Protocol_Description_STE.pdf

Key commits (not pushed)

Commit	Content
ba2a0e4	Coverage of record on the rev 1.1 RTL; RTM draft 2.0
57eca97	RTM draft 1.9 for spec rev 1.1
cec1f8d	Spec rev 1.1: registered Tx readback limits; ZT-04 scores both halves
27349f3	Spec rev 1.0: fractional fingerprint mean, 250 MHz ZCU106 build, TX-004 on silicon
bc740f9	ZCU106 PS DDR for the x16 SODIMM
e01b88e	Reference design built from the IP-XACT package
55b2a5f	Spec rev 0.8: product core meets 250 MHz after place and route
ee1a962	Spec rev 0.6: SECDDED state codes and watchdogs